

Ratul Hasan

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Professional Summary

Computer Science graduate with research experience in machine learning, healthcare AI, blockchain, and cybersecurity. Published in peer-reviewed conference proceedings, with research interests in trustworthy AI, large language models, and AI for healthcare. Aspiring to pursue graduate research in Artificial Intelligence.

EXPERIENCE

ELITE Research Lab LLC <i>Research Intern (Remote)</i> - Working on LLM regarding its bias, hallucination - Ongoing journal on LLM’s performance in mental health and depression	Queens, NY, USA May 2026 - Present
BAUET Computer Society, Department of CSE, BAUET <i>Programming Instructor</i> - Conducted regular training sessions on C programming and competitive programming for undergraduate students. - Designed lecture materials, coding exercises, and problem-solving sessions covering algorithms, data structures, and problem-solving techniques. - Mentored students in programming fundamentals and competitive programming, fostering analytical thinking and coding proficiency.	Natore, Bangladesh July 2024 - December 2025
Lab AR <i>Software Engineer Intern</i> - Led a team project to develop a web-based Property Management System. - Contributed to backend development using Laravel (PHP) and collaborated on feature implementation.	Dhaka, Bangladesh July 2025 - August 2025

EDUCATION

Bangladesh Army University of Engineering & Technology (BAUET) <i>B.Sc in Computer Science and Engineering CGPA: 3.82/4.0 Merit Position: 3rd</i> <i>Thesis Title: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer’s Disease</i>	Natore, Bangladesh 2022 – 2026
Govt. Majid Memorial City College <i>Higher Secondary School Certificate GPA: 5.0/5.0</i>	Khulna, Bangladesh 2019 – 2021

PUBLICATIONS

Conference Publications

Quantum-Resistant FOTA: End-to-End Decentralized Firmware Updates for IoT Using Blockchain and CRYSTALS-Dilithium	December 2025
Ratul Hasan , Md. Momenul Haque, Redoanul Haque, Yousuf Oley, and Ruhani Akter. <i>Proceedings of the 12th International Conference on Next Generation Computing, Communication, Systems and Security (NSysS ’25)</i> . ACM, New York, NY, USA, 110–114. doi.org/10.1145/3777555.3777569	
A Gas-Optimized Blockchain Framework for Scalable Document Verification Using Dynamic Access Control and IPFS	Expected 2026
Ratul Hasan , Samiha Farjana, Yousuf Oley, Md. Ohiduzaman Pranto, and Md Arik Rayhan. Accepted to the <i>International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)</i> . To be published in IEEE Proceedings.	
SplitSafe Ultra: A Moving-Target Defense Framework for Entropy-Aware Encrypted File Persistence on IPFS	Expected 2026
Md. Ahnaf Muhaimin, Ratul Hasan , Yousuf Oley, Samiha Farjana, Bristi Rani Roy, Subrata Kumer Paul, and Md. Ekramul Hamid. Accepted to the <i>5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)</i> . To be published in IEEE Proceedings.	
ActiVisionNet-Fusion: A Dual-Stream Explainable Deep Learning Framework for Driver Drowsiness Detection	Expected 2026
Nasim Ahmed, Ratul Hasan , Subrata Kumer Paul, Dewan Nafiul Islam Noor, Md. Momenul Haque, and Md. Ekramul Hamid. Accepted to the <i>5th IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON 2026)</i> . To be published in IEEE Proceedings.	

Journal Publications

BI-RNN: Biology-Informed Recurrent Neural Networks for Longitudinal Hippocampal Atrophy Forecasting in Alzheimer’s Disease	Under Review
Ratul Hasan , Md. Momenul Haque, Ananya Sarker. Under review in <i>Nature Scientific Reports</i> , Submission Date: May 2026.	

PROJECTS

LedgerSeal: A Blockchain Based Document Verification Framework using Dynamic Access Control and IPFS

[Github](#) 

- Developed a decentralized application (DApp) for constant-time document verification with dynamic access control on IPFS using a public blockchain.
- Tools Used: React JS, Anvil EVM, Foundry, Metamask, Pinata IPFS

BAUET Project & Thesis Archive

[Github](#) 

- Developed a repository website using Django(Both MVT & REST) where theses and projects of BAUET are represented
- Tools Used: HTML, Tailwind CSS, JavaScript, Django, PostgreSQL

ACHIEVEMENTS & CO-CURRICULAR ACTIVITIES

Programming Contests

- 1X ICPC Regionalist (Team: BAUET_Rong_Answer)
- 1X National Collegiate Programming Contest by JU (Rank: 149, Team: BAUET_PARADIGM)
- 2X Inter University Programming Contest (Best Rank: 45 at IUT, Team: BAUET_Twisted_Minds)
- Ranked 10th in Independence Day Programming Contest by Military Institute of Science & Technology
- 1X Champion, 1X 2nd runners-up in Intra University Programming Contest (Team)
- 1X Runners-up in Intra University Programming Contest (Solo)

Others

- 1X 1st Runners Up in Intra University Three Minutes Thesis Presentation

Online Judges

- Codeforces: Max Rating 1650 (Expert)
- Codechef: Max Rating 1676 (3 Star)
- Leetcode: Solved more than 50+ problems
- Solved more than 1000+ problems across various online judges

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TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL || **Frameworks:** Django, REST, Foundry, Laravel

Project Management: Git, Github || **Database:** MySQL, PostgreSQL

Technologies: Machine Learning, Computer Vision, Backend Development, Blockchain

VOLUNTEERING & LEADERSHIP EXPERIENCE

- President, BAUET Computer Society (Advanced through five executive leadership positions)
- Problem Setter, Intra-University Programming Contest, BAUET
- Officer of Documentation, Hult Prize at BAUET

REFERENCES

Prof. Mohammed Golam Sarwar Bhuyan

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